Homework #47

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Problem 3

Problem: $5y' + y = 0$

$$
\begin{aligned}
&y = \sum_{n=0}^{\infty} a_n x^n \\
&y' = \sum_{n=1}^{\infty} n a_n x^{n-1} \\
&5\sum_{n=1}^{\infty} n a_n x^{n-1} + \sum_{n=0}^{\infty} a_n x^n = 0 \\
&5\sum_{n=0}^{\infty} (n+1) a_{n+1} x^n + \sum_{n=0}^{\infty} a_n x^n = 0 \\
&5\sum_{n=0}^{\infty} [(n+1) a_{n+1} + a_n] x^n = 0.
\end{aligned}
$$

Thus,

$$(n+1)a_{n+1} + a_n = 0$$

Solving for a_{n+1}:

$$a_{n+1} = -\frac{a_n}{n+1}$$

Creating a table:

n	$\mathbf{a_n}$
0	a_0
1	$-a_0$
2	$\frac{a_0}{2}$
3	$-\frac{a_0}{3\cdot 2}$

Thus, the explicit formula is

$$a_n = (-1)^n \frac{a_0}{n!}$$

And solving for y results in

$$\begin{aligned} y &= \sum_{n=0}^{\infty} (-1)^n \frac{a_0}{n!} x^n \\ &= a_0 \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!} \end{aligned}$$

Answer: $\boxed{y = a_0 \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}}$

Problem 4

Problem: $(x + 2)y' + y = 0$